

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Cummins® electronic service tool or equivalent.

Additional Service Items

- No additional service items required.

General Information

The Aftertreatment Diesel Particulate Filter (DPF) Regeneration Analyzer Test is a diagnostic used to identify malfunctioning engine performance components. The test is located in INSITE™ electronic service tool under the Diagnostic Tests tab.



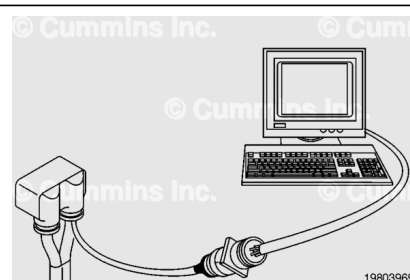
The test status will be shown in the status window.

1. Test description window
2. Instructions window
3. Status window
4. Status bar - shows progress of the test (will disappear when the test is complete).

System Requirements

The Aftertreatment Diesel Particulate Filter Regeneration Analyzer Test requires:

- INSITE™ electronic service tool version 8.5.2 or later.
- Minimum of 250 MB of available computer hard drive space before starting the test.



The Aftertreatment Diesel Particulate Filter Regeneration Analyzer Test is **only** to be used when directed by published troubleshooting.

Preparatory Steps

WARNING

During testing, exhaust gas temperature could reach 800°C [1500°F] and exhaust system surface temperature can exceed 700°C [1300°F], which is hot enough to ignite or melt common materials, and to burn people. The exhaust and exhaust components can remain hot after the vehicle stops moving. To avoid the risk of fire, property damage, burns or other serious personal injury, allow the exhaust system to cool before beginning this procedure or repair and make sure that no combustible materials are located where they are likely to come in contact with hot exhaust or exhaust components.

Before the Aftertreatment Diesel Particulate Filter Regeneration Analyzer Test, follow the steps listed below:

1. Select an appropriate location to park the vehicle.
 - a. On a surface that will **not** burn or melt under high temperatures (such as clean concrete or gravel, **not** grass or asphalt)
 - b. Away from anything that can burn, melt, or explode
 - Nothing within 0.6 m [2 ft] of the exhaust outlet
 - Nothing that can burn, melt, or explode within 1.5 m [5 ft] (such as gasoline, wood, paper, plastics, fabric, compressed gas containers, and hydraulic lines)
 - No gas or vapors nearby that could burn, explode, or contribute to a fire (such as liquid petroleum gas, gasoline vapors, oxygen, and nitrous oxide).
2. Park the truck securely.
 - a. Set the parking brake.
 - b. Place the transmission in Park, if provided. Otherwise, place the transmission in Neutral.
 - c. Set the wheel chocks at the front and rear of at least one tire.
3. Set up a safe exhaust area.
 - a. If bystanders can possibly enter the area, use barriers to keep people at least 1.5 m [5 ft] from the exhaust outlet during regeneration.
 - b. When indoors, attach an exhaust discharge pipe rated for at least 800°C [1500°F].
 - c. Keep a fire extinguisher nearby.
4. Check exhaust system surfaces.
 - a. Confirm that nothing is on or near the exhaust system surfaces (such as tools, rags, grease, or debris).
5. Prepare for engine speed changes during regeneration.

- a. Do **not** operate **any** power takeoff (PTO)-powered devices. Disconnect these devices before starting regeneration.
 - b. Stay clear of the engine compartment.
6. To begin the test:
- a. The engine will create enough heat to regenerate the aftertreatment DPF. Engine speed can increase and the turbocharger can whistle loudly during the regeneration process. Once the test completes, the engine will automatically return to normal idle speed.
 - b. Verify the vehicle and surrounding area is monitored during the test. If any unsafe condition occurs, shut the engine off immediately.

Once the Aftertreatment Diesel Particulate Filter Regeneration Analyzer Test is complete, exhaust gas and exhaust surface temperatures will remain elevated for 3 to 5 minutes. Keep the engine running at idle until the exhaust temperatures are reduced.

Test

Note : If the connection between INSITE™ electronic service tool and the engine control module (ECM) is lost for any reason, a pop-up message will appear. The test can be restarted after cycling the keyswitch OFF for 90 seconds and then back ON.

1. Begin the Aftertreatment Diesel Particulate Filter Regeneration Analyzer Test.
 - Use INSITE™ electronic service tool to perform the test.
 - Engine speed will increase to a specified value. Then the ECM will perform the analysis process.
 - At the conclusion of the test, the engine will return to idle. The Aftertreatment Diesel Particulate Filter Regeneration Analyzer Test will display either a Pass or Fail status for all systems.
2. Monitor the area.
 - Verify the vehicle and surrounding area are monitored during the test. If any unsafe condition occurs, shut the engine off immediately.
 - To stop the test, click the STOP button on the test screen. The test can also be stopped by engaging the clutch, brake, or throttle pedal; or by turning the engine off.
 - Test progress can be monitored using the status bar.
 - Test results will be displayed after the test completes and analyzes.
 - Once the test is complete, the engine will automatically return to normal idle speed. The test will display the test results for review until the next key cycle.

A logfile will be automatically created and saved to the computer. You may be asked to provide the logfile by Cummins® CARE if technical assistance is required.

The Aftertreatment Diesel Particulate Filter Regeneration Analyzer Test will **not** start or will be aborted if:

- Accelerator pedal is depressed

- Clutch pedal is depressed
- Brake pedal is depressed
- Parking brake **not** set
- Transmission is put into gear
- PTO engaged
- Vehicle speed detected
- Engine protection state active
- Regeneration inhibiting fault code becomes active
- High aftertreatment temperature faults become active.

If the Aftertreatment Diesel Particulate Filter Regeneration Analyzer Test aborts or will **not** activate, a message will be displayed. Correct the issue identified before proceeding. For more information on abort messages and associated repair action, see the Troubleshooting section of this procedure.

Troubleshooting

This section is used to assist troubleshooting abort messages from the Status Window.

Abort messages will be displayed in the Status Window. The most recent message will appear at the bottom.



Status Window Message	Action
The test was stopped as there was not enough valid data.	- Inspect for exhaust manifold leaks. Refer to Procedure 010-024 in Section 10 (/qs3/pubsys2/xml/en/procedures/10/10-010-024-tr.html). - Clean and inspect the exhaust gas recirculation (EGR) valve. Refer to Procedure 011-022 in Section 11 (/qs3/pubsys2/xml/en/procedures/10/10-011-022-tr.html). - Check for aftertreatment diesel oxidation catalyst (DOC) face plugging. Clean and inspect for reuse. Refer to Procedure 011-049 in Section 11 (/qs3/pubsys2/xml/en/procedures/10/101-011-049-tr.html). - Clean and inspect the aftertreatment fuel injector. Refer to Procedure 011-042 in Section 11 (/qs3/pubsys2/xml/en/procedures/10/101-011-042-tr.html). - Perform the Aftertreatment Fuel Injector Flow Test. Refer to Procedure 011-054 in Section 11 (/qs3/pubsys2/xml/en/procedures/10/101-011-054-tr.html). - Inspect the cooling fan for damage. Refer to Procedure 008-040 in Section 8 (/qs3/pubsys2/xml/en/procedures/10/10-008-040.html). - Check the coolant thermostat. Refer to Procedure 008-013 in Section 8 (/qs3/pubsys2/xml/en/procedures/10/10-008-013-tr.html).
There is not enough available storage space. A minimum of 250 MB is required.	Minimum of 250 MB memory space is required to run this test.
INSITE™ electronic service tool can not currently read the ECM parameters from the connected ECM. Contact technical support.	Follow local escalation process.
Data files are not available. Proceed with published troubleshooting.	Proceed with published troubleshooting.

Status Window Message	Action
The user does not have sufficient access permissions to create the output file. Contact the local IT administrator.	• Contact the local IT administrator.
The test has completed and failed.	• Read and record status message.
The test has stopped or could not start because the data link connection was lost.	• Check the cables between the computer data link adapter for proper connection and condition. Refer to Procedure 019-165 in Section 19 (/qs3/pubsys2/xml/en/procedures/99/99-019-165.html) of the Signature™, ISX, and QSX15 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666259.

Finishing Steps

- Do not turn the keyswitch off until the test has completed and results are displayed.
- Check for any active fault codes. If active fault codes are present, follow published troubleshooting.